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Prevalence of low back pain and its associated factors among weavers in low- and middle-income countries: a systematic review and meta-analysis

Abebe Kassa Geto^{1*}, Leykun Berhanu², Gete Berihun³, Chala Daba² and Belay Desye²

Abstract

Introduction Low back pain, one of the musculoskeletal disorders is among the major global public health problems and contributors to disability and workers' absence in occupational areas which certainly disrupts work productivity and the expected results. Though various studies investigated low back pain, the results were inconsistent and inconclusive enough, and there is no representative data in low- and middle-income countries on this public health concern. This in turn hinders the efforts of various intervention activities. Therefore, this systematic review and meta-analysis was conducted to determine the pooled prevalence of low back pain and its associated factors among weavers of low- and middle-income countries.

Methods All the relevant articles were retrieved from databases such as PubMed/MEDLINE, CINAHL, LIVIVO, African Journals Online, African Index Medicus (AIM), HINARI, Science Direct, Web of Science, Cochrane Library, Google Scholar, Semantic Scholar, and Google. The Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guideline was followed for this study. The extracted data were analyzed using STATA 17 software. With a 95% confidence interval, this meta-analysis with the random-effects model was conducted to determine the pooled prevalence.

Result A total of twenty articles with 7211 study participants were included in this meta-analysis. The pooled prevalence of low back pain was 55.81%. Age, working in a chair with no back support, working in an uncomfortable posture, work experience, and job stress were the factors significantly associated with low back pain.

Conclusion A high prevalence of low back pain among weavers in low- and middle-income countries was registered. This indicates the need to take effective intervention measures. Rigorous provision of ergonomic training, providing lengthy breaks, improving workplace ergonomic design, and increasing job satisfaction are recommended.

Keywords Low back pain, Low- and middle-income countries, Job stress, Prevalence, Weavers

*Correspondence:

Abebe Kassa Geto
abebekassa2129@gmail.com

¹Department of Public Health, College of Health Sciences, Woldia University, Woldia, Ethiopia

²Department of Environmental Health, College of Medicine and Health Sciences, Wollo University, Dessie, Ethiopia

³Department of Environmental Health, College of Medicine and Health Sciences, Debre Markos University, Debre Markos, Ethiopia



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Introduction

Low back pain (LBP) is a common condition characterized by discomfort or pain localized to an area extending from the twelfth (12th) rib, the last bony prominence on the back of the rib cage, down to the inferior gluteal folds, the creases at the bottom of the buttocks [1]. It is the most common musculoskeletal disorder (MSD) affecting the working population [2, 3]. The potential sources of LBP symptoms are diverse, ranging from nerve roots and muscles to bones, joints, intervertebral discs, and even abdominal organs [3–5]. LBP is the most frequently reported musculoskeletal issue and imposes a significant burden on individuals, healthcare systems, and social care services, with indirect costs being the most substantial [6].

LBP along with other MSDs in the working environment are a major public health concern across the globe. They are a very common occupational area health problem mostly manifested by a range of symptoms like pain, aches, and discomfort in the body.

LBP has been stated as one of the major contributors to disability and workers' absence in the workplace which certainly disrupts work productivity and expected work results. It is also responsible for multiple work interruptions/stoppages and substantial direct and indirect costs [7–10]. Workers of many occupations live with a health burden linked with the disabling musculoskeletal pain and injuries of an occupational-related cause [11].

According to a recent Global Burden of Disease (GBD) report, years lived with disability (YLDs) accounted for 31.3% of the global disability-adjusted life years (DALYs) in 2021. From this, with 70.2 million YLDs, LBP was the leading Level 3 cause of YLDs in the world [12]. Despite many of the work-related problems are preventable, there are about 2.9 million work-related deaths globally every year. Moreover, globally 6% of all the deaths were attributed to be work-related. More than 7,500 people die following workplace accidents in every single day [13].

A systematic review and meta-analysis on the global prevalence of work-related musculoskeletal disorders among physiotherapists found that WMSD pooled prevalence at the lower back was 40.1% [7]. Another systematic review and meta-analysis conducted among African school teachers revealed that the estimated pooled prevalence of LBP was found to be 59% [14]. Moreover, a systematic review and meta-analysis done among nursing and medical students found that the pooled prevalence rates of LBP for nursing and medical students were 44% and 53% respectively [15]. According to the systematic review and meta-analysis conducted in Ethiopia, the overall pooled annual prevalence of LBP was estimated to be 54.05% [16]. A pooled LBP prevalence rate of 65.0% was recorded according to a systematic review and meta-analysis among health workers in Saudi Arabia [17].

Weaving is one of the globe's oldest surviving crafts and a valuable cottage industry in developing nations including India, Iran, Turkey, China, Bangladesh and Pakistan where traditional weaving methods are still commonly practiced [18]. Weaving sustains millions of jobs, particularly in rural areas, by supporting artisans, small-scale producers, and local enterprises, contributing to economic stability and foreign exchange earnings in many LMICs. Additionally, traditional weaving strengthens cultural identity and tourism while promoting agricultural industries through the use of locally sourced materials [19]. A handloom is a wooden and iron device used for weaving cloth without the use of electric motors, relying instead on the weavers to move the fabric manually with their hands and feet [20, 21]. Weaving requires various tasks that are carried out by continuously sitting in a fixed position to create fabric and by repetitively moving the upper and lower limbs to manipulate pedals and shuttles, with the arms extended away from the body [22]. Performing activities strenuously and for lengthy working hours, inability to take regular physical exercise and illiteracy make weavers among the populations highly at risk of WMSDs such as LBP [18, 23]. Weaving requires repetitive motions that strain the musculoskeletal system, heightening the risk of fatigue and limiting the chance for tissues to recover, which leads to pain and discomfort [24]. LBP is a multi-factorial problem having many possible causes and determining its predictors is a very difficult task. Factors associated LBP include age, gender, body mass index (BMI), educational background, work experience, working hours, lack of safety training, awkward working postures, work shifts, prolonged standing, lifting heavy objects, sleep disturbances, and previous medical history of musculoskeletal disorders [16, 18, 25–30]. Moreover, LBP and other WMSDs were significantly associated with the absence of breaks, working longer than eight hours a day, working seat with no back support, repetitive movement, and working in a static position. Furthermore, LBP and other work-related musculoskeletal disorders were strongly linked to the lack of breaks, working more than eight hours daily, using seats without back support, repetitive movements, and maintaining static positions [18, 31–34]. Psychological and behavioral factors such as alcohol, smoking, job dissatisfaction, job stress, limited social support from workplace partners and supervisors, sleeplessness, and depression were also found to be associated with LBP [18, 32, 35–38].

The prevalence of LBP remains less focused and overlooked in low- and middle- income countries (LMICs) due to the focus on more pressing and life-threatening health issues like infectious diseases and non-communicable diseases (NCDs) [39]. Although many primary studies have been carried out on LBP among weavers across various LMICs of the world, the findings have

not been consistent with the prevalence ranging from 15.2% [40] to 82.91% [41] and conclusive, which could hamper the assessments of ongoing intervention efforts and activities. Moreover, no study provides evidence on the pooled prevalence of LBP among weavers in LMICs. Therefore, this systematic review and meta-analysis aimed to estimate the pooled prevalence of LBP and identify its associated factors among weavers in LMICs. The results of this systematic review and meta-analysis are used to emphasize the need for primary prevention of LBP by promoting the health of weavers and the economic development of the countries. Furthermore, the results will help policy and decision-makers, healthcare officials, and other bodies to plan and implement strategies to prevent, control, and develop a safe working environment to reduce the impacts of LBP among weavers. Moreover, the findings will aid policymakers, healthcare planners, and other relevant stakeholders in designing and implementing strategies to prevent and control LBP among weavers and to create a safer working environment to mitigate its impact.

Materials and methods

Protocol registration

The protocol for this systematic review and meta-analysis has been registered on the International Prospective Register of Systematic Reviews (PROSPERO). CRD42024567386 is its registration number.

Search strategy and study selection

In this systematic review and meta-analysis, published and unpublished studies were searched from different electronic databases such as PubMed/MEDLINE, Science Direct, Cochrane Library, LIVIVO, CINAHL, African Journals Online, Web of Science, African Index Medicus (AIM), HINARI, Semantic Scholar, Google, and Google Scholar. Besides, gray literature was also identified from digital libraries and repositories of thirteen different universities. The search was carried out using various keywords combined with the appropriate Boolean operators: “AND” or “OR” (Supplementary material 1). The search was performed up to 21 June 2024 by four independent authors (AKG, BD, GB, and LB). The articles searched from the selected electronic databases were transferred to the Endnote version 8 software and all duplicate files were excluded. Selecting all the relevant articles followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines [42] (Supplementary material 2).

Inclusion criteria

Population

All the studies conducted on LBP among weavers in LMICs.

Exposure

Weavers that exhibited LBP.

Comparison

Weavers that didn't exhibit LBP.

Outcome

Studies assessed low back pain as a primary outcome in a twelve-months period.

Study setting

Studies conducted in community as well as institutions.

Study design

All studies following cross-sectional, cohort, and case-control study designs were included.

Publication

Published and unpublished studies.

Country

All the relevant studies conducted in low- and middle-income countries.

Language

Studies reported only in English language were included.

Exclusion criteria

Studies with no full text, unidentified reports, abstracts, editorials, irretrievable studies, letters, qualitative studies, and studies that did not report the outcome of interest (LBP) were excluded from this systematic review and meta-analysis.

Outcome assessment

The main outcome of this study was to determine the pooled prevalence of LBP among weavers in low- and middle-income countries. This is estimated by multiplying by 100 after the number of study participants (the numerator) who had LBP was divided by the actual sample size (the denominator). Apart from this, the systematic review and meta-analysis was conducted to identify the factors associated with LBP in the form of log odds ratio.

Operational definition

Low back pain is a common condition characterized by discomfort or pain localized to an area extending from the twelfth (12th) rib, the last bony prominence on the back of the rib cage, down to the inferior gluteal folds, the creases at the bottom of the buttocks [1].

Data extraction procedure, study quality, and risk of bias assessment

A standard data extraction template consisting of various details of the study such as author name, year of publication, country, continent, study design, article quality score, weaving type, data collection tools, methods of data collection, sample size, and prevalence was prepared. Duplicate articles were removed after the relevant articles for inclusion were carefully screened by three different reviewers (AKG, LB, and TF). Four independent authors (AKG, BD, CD, and LB) undertook all the required data extraction activities. Using the Joana Briggs Institute (JBI) checklist of critical appraisal for cross-sectional studies, the quality of each article was critically evaluated [43] (Supplementary material 3). With scores measured on a scale of 100%, the quality of each article was independently assessed by three different authors (AKG, BD and GB). For further analysis, articles having a quality score of above 50% were included [44, 45]. A mean score was calculated from all the reviewers' evaluations to address and resolve any discrepancies encountered during the quality assessment.

Statistical analysis procedures

After the data were extracted, analysis was made using STATA (Corporation, College Station, Texas, USA) version 17 software. Heterogeneity within the included studies was assessed using the Higgs I^2 test, with values of 75%, 50%, and 25% showing high, moderate, and low levels of heterogeneity respectively [46]. In this meta-analysis, high heterogeneity was observed across the included studies ($I^2 = 97.93\%$; $p = 0.00$). With a 95% confidence interval, a restricted maximum-likelihood [47] method of random-effects model was used to determine the pooled prevalence of LBP among weavers. The odds ratio was computed to show the strength of the association between LBP (the outcome variable) among weavers and its associated factors.

To present the pooled prevalence of LBP, a forest plot was used. To determine the influence of an individual study on the pooled prevalence estimate of LBP, a clear sensitivity analysis was performed. Sub-group analysis was also done to identify the possible sources of heterogeneity based on the year of publication (before 2020 and 2020 and after), study country category (Ethiopia, India, Iran, and other countries), continent (Africa and Asia), weaving type (cloth and carpet), and method of data collection (interview only, interview and observation in combination, and self-administration). Besides, the presence of potential publication bias was determined by using a funnel plot and Egger's test [48].

Results

Study selection

A total of 941 studies were identified from an electronic database and reference searching. Endnote 8 was used as a reference manager. One hundred and sixty-two duplicated articles were removed. The total number of articles excluded based on their titles and abstracts due to the failure to meet the criteria of inclusion was 741. Moreover, thirteen articles were removed as they did not report the outcome of interest and five articles were excluded as they failed to meet quality assessment methods. In this meta-analysis, a total of 20 full-text articles were included to estimate the pooled prevalence of LBP among the weavers of low-and middle- income countries by following the PRISMA guideline (Fig. 1).

Characteristics of the included studies

This meta-analysis included twenty cross-sectional studies, encompassing a total of 7,211 participants. The highest prevalence of LBP among weavers was found to be 82.91% [41] and the lowest prevalence was 15.2% [40] among the included studies. Regarding the continent, fourteen of the included studies were conducted in Asia [23, 25, 26, 31, 49–58], and the remaining six studies were conducted in Africa [34, 40, 59–62]. Regarding the country in which the studies were conducted, seven were in India [25, 26, 31, 50, 54, 56, 57], five studies were in Ethiopia [34, 40, 59, 62], four were in Iran [51–53, 58], two studies were in Bangladesh [23, 55], and the other two studies were conducted in Burkina Faso [60], and Thailand [49]. Regarding the data collection methods, thirteen of the included articles [26, 50–59, 62–64] [23, 49–57] used interviews as a single data collection method, five of the included studies [25, 26, 34, 40, 58, 59] used interview and observation in combination, and the other two studies [25, 63] used self-administration (Table 1).

Meta-analysis

Pooled prevalence of low back pain among weavers

Twenty articles were included to determine the pooled prevalence of LBP in in this meta-analysis. The pooled prevalence of LBP among weavers in low-and middle-income countries was found to be 55.81% (95% CI: 48.18%, 63.44%). A random effects model was employed to estimate the pooled prevalence of LBP following the high level of heterogeneity among the included studies ($I^2 = 97.93\%$; $p = 0.00$) (Fig. 2).

Test for publication bias

The funnel plot indicated that there is no significant publication bias (Fig. 3). Statistically, Eggers' test result also depicted the absence of statistically significant publication bias (small studies effect) ($p = 0.161$).

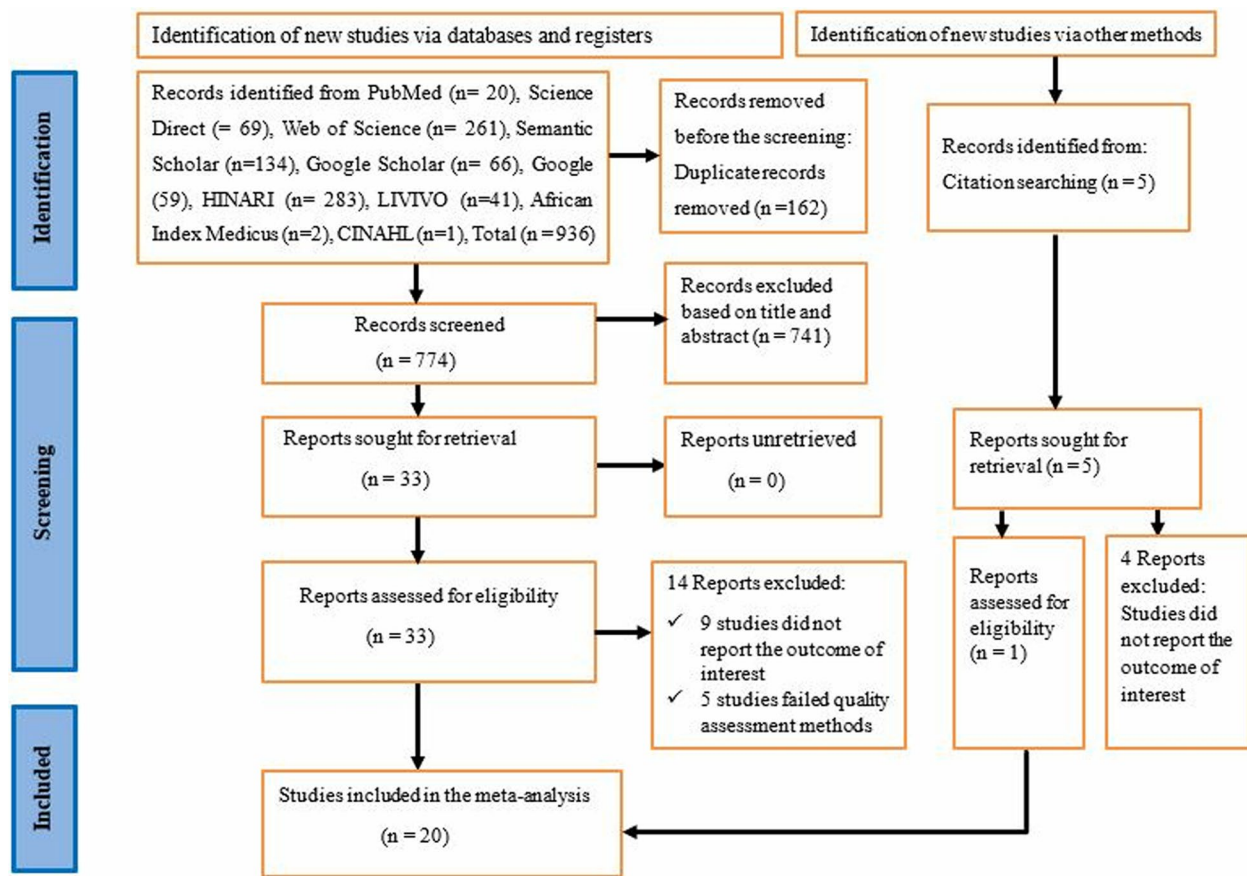


Fig. 1 A PRISMA flow diagram showing study selection for systematic review and meta-analysis on the prevalence of LBP and its associated factors among weavers in low- and middle-income countries, 2024

Sensitivity analysis

The effect of individual studies on the pooled estimate of LBP was evaluated by performing a sensitivity test. The finding revealed that none of the included studies had an effect on the pooled estimate (Fig. 4).

Subgroup analysis

Based on the continents in which the studies have been conducted, the highest pooled prevalence of LBP was recorded in Asia (58.04%, 95% CI: 49.44–66.64%) as compared to studies conducted in Africa (50.75%, 95% CI: 34.69–66.81%) (Fig. 5). Based on the country category in which the studies were conducted, the highest LBP pooled prevalence was registered in India (67.40%, 95% CI: 59.13–75.67%) followed by Iran (52.34%, 95% CI: 41.90–62.79%) (Fig. 6). Regarding the articles' publication year category, the highest pooled LBP was observed among studies published in 2020 and after (56.45%, 95% CI: 47.51–65.39%) as compared to those studies published before 2020 (54.38%, 95% CI: 38.56–70.20%) (Fig. 7).

Based on the variable weaving type, the highest pooled LBP prevalence was recorded among studies conducted

on cloth weavers (56.56%, 95% CI: 46.05–67.07%) of low- and middle-income countries as compared to carpet weavers (54.05%, 95% CI: 46.39–61.72%) (Fig. 8). The highest pooled prevalence was recorded for those studies that used self-administration (70.45%, 95% CI: 53.39–87.52%) as compared to studies that used interview alone as a data collection method (58.69%, 95% CI: 49.98–67.40%) and a combination of interview and observation (42.75%, 95% CI: 28.11–57.38%) (Fig. 9). Based on sample size category, the highest pooled LBP prevalence was observed among studies having a sample size of lower than 361 (59.77%, 95% CI: 50.55–69.00%) as compared to studies having a sample size of 361 and above (48.66%, 95% CI: 36.07–61.24%) (Fig. 10).

Meta-regression

To identify the source of heterogeneity by considering continent, country category, publication year category, weaving type, the method of data collection, and sample size category as factors, a univariate meta-regression analysis was performed. However, none of the variables demonstrated statistical significance (Table 2).

Table 1 Characteristics of the included studies to determine the pooled prevalence of LBP among weavers of low-and middle - income countries, 2024

Authors	Publication Year	Country	Continent	Study Design	Method of data collection	Weaving Type	Sample Size	Prevalence (%)	Quality Score (%)
Terfe et al [59]	2023	Ethiopia	Africa	Cross-sectional	Interview and observation	Cloth	660	44.1	87.5
Choobineh et al [64]	2007	Iran	Asia	Cross-sectional	Interview and observation	Carpet	1439	81	75
Haftu et al [40]	2023	Ethiopia	Africa	Cross-sectional	Interview and observation	Cloth	420	46.9	100
Durlov et al [57]	2014	India	Asia	Cross-sectional	Interview	Cloth	175	68	87.5
Gada et al [56]	2024	India	Asia	Case control	Interview	Carpet	200	51.5	62.5
Tefera Zele et al [34]	2020	Ethiopia	Africa	Cross-sectional	Interview and observation	Cloth	814	48.9	100
Hossain et al [55]	2018	Bangladesh	Asia	Cross-sectional	Interview	Cloth	230	27.4	62.5
Jamil et al [23]	2022	Bangladesh	Asia	Cross-sectional	Interview	Cloth	250	82	75
Kaboré and Schepens [60]	2023	Burkina Faso	Africa	Cross-sectional	Interview	Cloth	247	85.4	100
Laeek Ahemad et al [41]	2021	India	Asia	Cross-sectional	Interview	Cloth	234	100	100
Naz et al [54]	2015	India	Asia	Cross-sectional	Interview	Cloth	64	76.56	62.5
Nazari [53]	2024	Iran	Asia	Cross-sectional	Interview	Carpet	240	83.5	75
Zinabu et al [61]	2024	Ethiopia	Africa	Cross-sectional	Interview	Cloth	414	76.3	100
Nazari et al [52]	2012	Iran	Asia	Cross-sectional	Interview	Carpet	200	100	62.5
Pavana [63]	2021	India	Asia	Cross-sectional	Self-administration	Cloth	221	79.2	87.5
Satheeshkumar and Krishnakumar [25]	2020	India	Asia	Cross-sectional	Self-administration	Cloth	361	61.77	87.5
Telaprolu and Anne [26]	2014	India	Asia	Cross-sectional	Interview and Observation	Cloth	227	375	75
Thongsuk and Geater [49]	2021	Thailand	Asia	Cross-sectional	Interview	Carpet	163	96.3	62.5
Nejad et al [51]	2021	Iran	Asia	Cross-sectional	Interview	Carpet	91	90.1	75
Zinabu et al [62]	2024	Ethiopia	Africa	Cross-sectional	Interview	Cloth	403	63.5	100

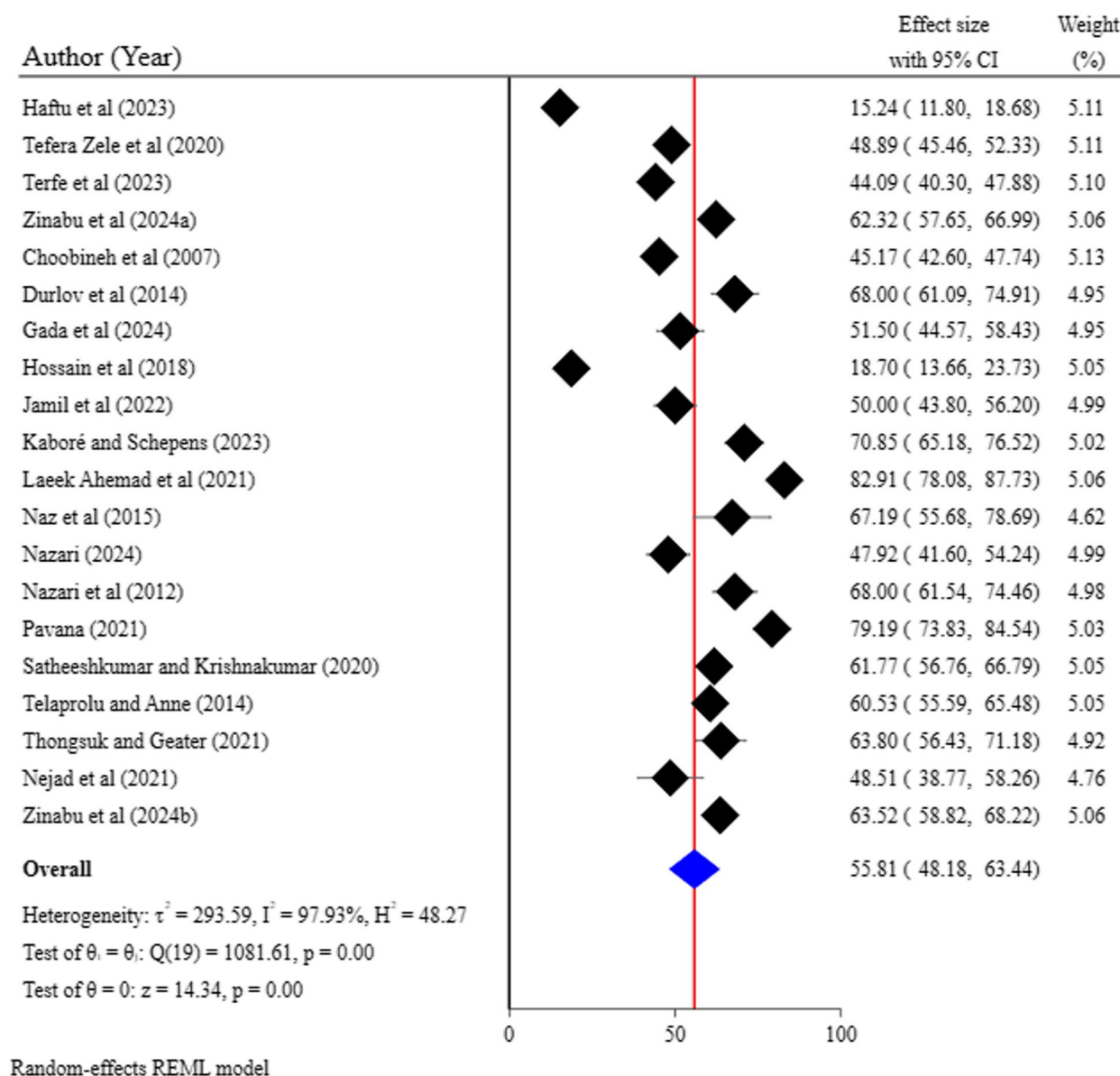


Fig. 2 A forest plot showing the pooled prevalence of LBP among weavers in low-and middle-income countries, 2024

Factors associated with low back pain among weavers of LMICs

Seven factors were repeatedly presented in the included articles of this meta-analysis. The factors were gender, work experience, working hours per day, working in a chair with no back support, working in an uncomfortable posture, and job stress. However, five of the factors (age > 40 years, working in a chair with no back support, working in an uncomfortable posture, work experience (> 10 years) and presence of job stress) were found to be significantly associated with the pooled prevalence of LBP among weavers of low-and middle-income countries (Table 3).

The association between the age of the weavers and LBP among the included two studies [23, 60] has been

assessed. The result showed that there was a significant association in both of the included studies. According to this meta-analysis, the odds of LBP were 81% higher among weavers aged > 40 years as compared to those who were 40 years and below (POR = 1.81; 95% 1.06–2.56). The association between working in a chair with no back support and LBP among the included two studies [34, 62] has been determined. The result showed that there was a significant association in both of the studies included. According to the result of the random effect meta-analysis, the odds of LBP were 2.20 times higher among weavers working in a chair with no back support as compared to their counterparts (POR = 2.20; 95% CI: 1.54–2.87).

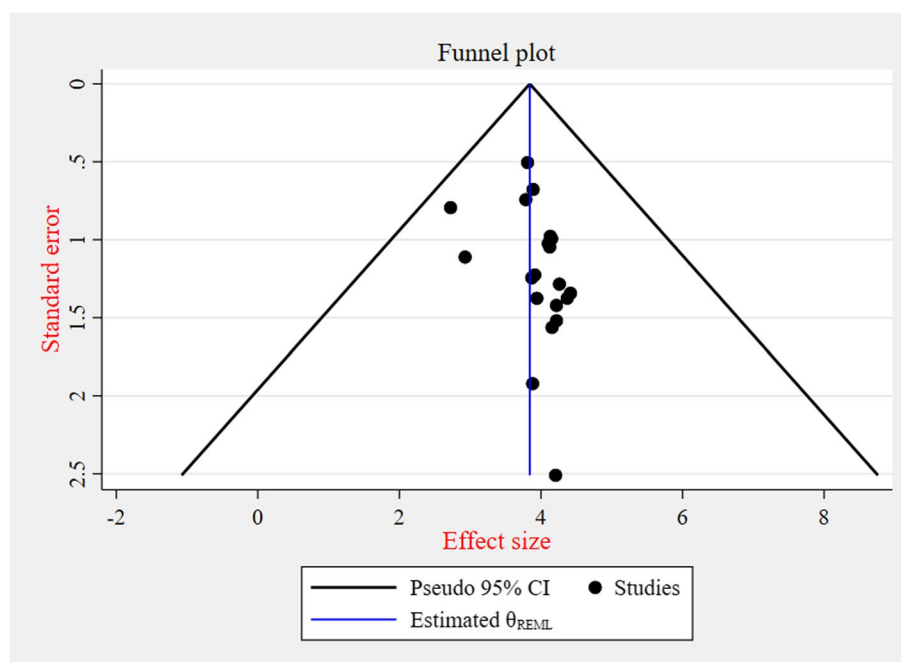


Fig. 3 A funnel plot to test the publication bias of the included studies of the meta-analysis

Based on the findings of two studies [26, 62], the association between LBP and working in an uncomfortable posture was assessed. In two of these studies [26, 62], a positive association was observed. According to the results of this meta-analysis, the odds of LBP were 2.57 times higher among weavers who worked in an uncomfortable posture as compared to those who worked in a comfortable posture (POR=2.57; 95% CI: 1.25–3.90) (Table 3). The association between work experience and prevalence of LBP was assessed by including two studies [59, 60]. One of the included studies [60] showed a positive association with LBP. The result of this meta-analysis found that the odds of LBP were 54% higher among weavers who had a work experience of > 10 years as compared to weavers having a work experience of 10 years and lower (POR=1.54; 95% CI: 1.17–1.91). Similarly, two articles [34, 62] were included to determine the association between job stress and LBP. In this case, both articles [34, 62] had a positive and significant association with LBP. This meta-analysis demonstrated 2.15-fold increased odds of LBP among weavers with job stress compared to those without (POR=2.15; 95% CI: 1.33–2.97) (Table 3).

Discussion

LBP has continued to be a global burden to have a substantial impact on the health as well as the productivity of workers in workplace settings. So far, many studies have been conducted to identify the potential risk factors associated with LBP. However, these findings were found to be inconsistent and inconclusive enough. This in turn

hinders the efforts of effective and timely intervention activities. To determine the prevalence of LBP and its associated risk factors among weavers in LMICs, a systematic review and meta-analysis was conducted.

The pooled prevalence of LBP among weavers in LMICs was found to be 55.81% (95% CI: 48.18%, 63.44%). This figure is in line with a global systematic review and meta-analysis pooled prevalence report on musculoskeletal disorders at the lower back among operating room personnel (61.48%) [65] and the pooled prevalence of LBP in Ethiopia (54.05%) [16]. Moreover, it is congruent with the prevalence at the back region of surgeons undertaking robotic- and video-assisted surgery (RVAS) (49.9%) [66], the prevalence of LBP among the African population (57%) [67], and the school teachers in Africa (59.0%) [14]. However, the figure is higher than a global systematic review and meta-analysis pooled prevalence report on musculoskeletal disorders at the lower back among physiotherapists (40.1%) [7] and nursing students (44%) [15]. The discrepancies might be due to the professional knowledge regarding LBP, degree of job rotation, provision of psychosocial training, workload reduction, and socio-economic matters. The odds of LBP were 81% higher among weavers aged > 40 years as compared to those who were 40 years and below. This is supported by a study conducted in Brazil in which the increase in age was associated with the prevalence of LBP [68].

The odds of LBP were 54% higher among weavers who had a work experience of > 10 years as compared to weavers having a work experience of 10 years and lower. The reason for this could be prolonged work, leading to

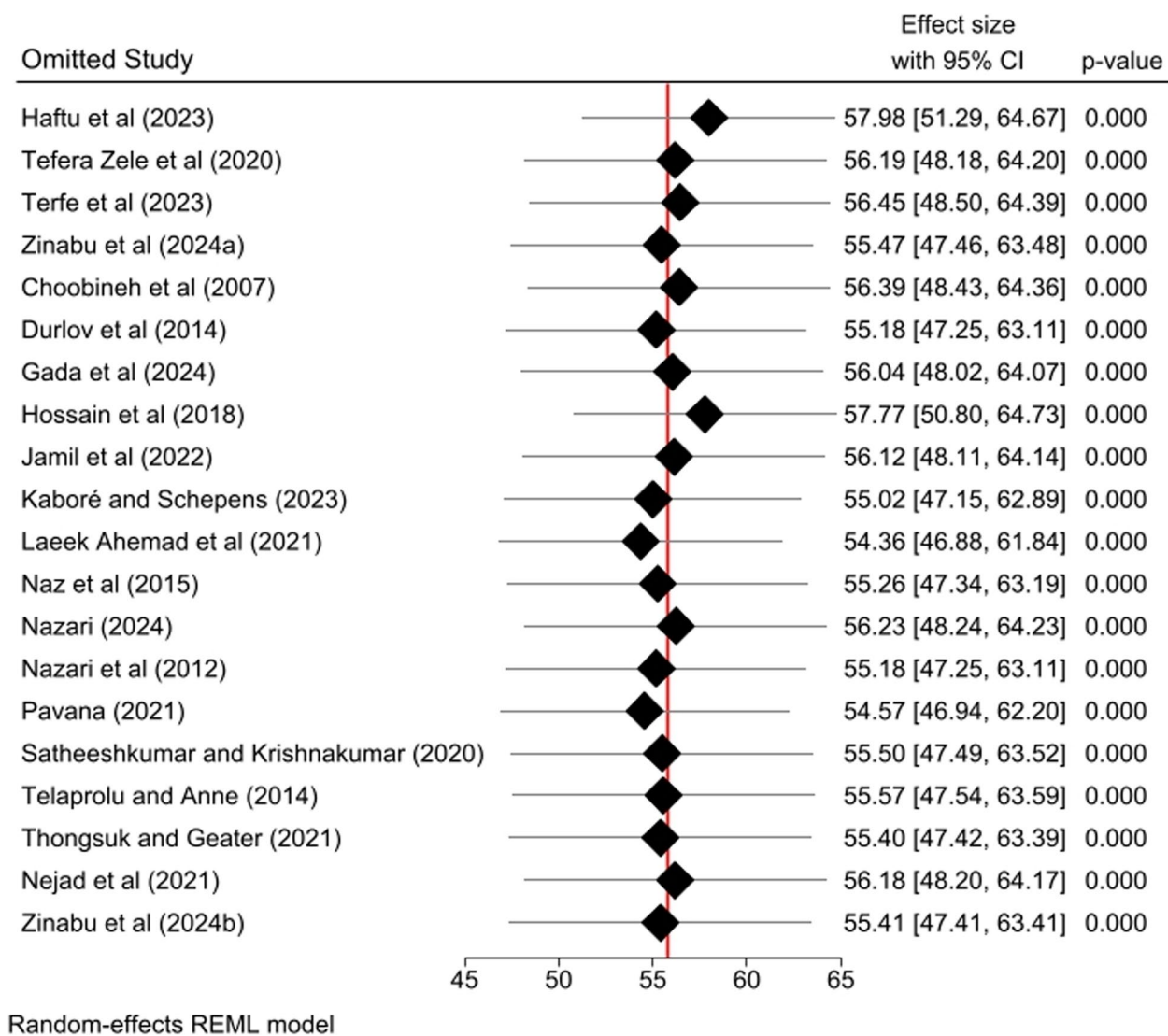


Fig. 4 A sensitivity analysis result of the included studies for LBP among weavers in low-and middle-income countries, 2024

cumulative trauma and repetitive strain injuries, combined with inadequate recovery time from fatigue and stress, can negatively impact the musculoskeletal system and cause LBP [69]. The finding is also supported by studies conducted in Korea and Ethiopia which elucidated that due to an increased working period, the risk of WMSDs such as LBP is severe and getting worse [70, 71].

The odds of LBP were 2.20 times higher among weavers working in a chair with no back support as compared to their counterparts. This finding is supported by a study in India on female textile workers in which workers sitting without back support were at higher risk of developing work-related musculoskeletal disorders [72] and an intervention study conducted on office workers evidenced that workers who received a highly adjustable chair for sitting significantly reduced musculoskeletal symptoms [73].

The odds of LBP were 2.57 times higher among weavers who worked in an uncomfortable posture as compared to those who worked in a comfortable posture. This is supported by evidence that shows working in awkward postures (twisted or bent backs) increases the risk of work-related musculoskeletal disorders, such as LBP, compared to maintaining a straight back [74–76]. The reason for this is, poor posture leads to stiffness, muscle compression, and subsequent pain and discomfort throughout the body [77]. Posture significantly influences the force generated during activities. Back bending, twisting, or bending of joints (shoulders, wrists, hips, knees) increases stress on muscles, joints, and nerves, causing fatigue and potentially injuries [78]. Moreover, WMSDs like LBP, are more prevalent with fixed postures than with varied counter working positions [76, 78, 79]. According to studies conducted in Addis Ababa [79]

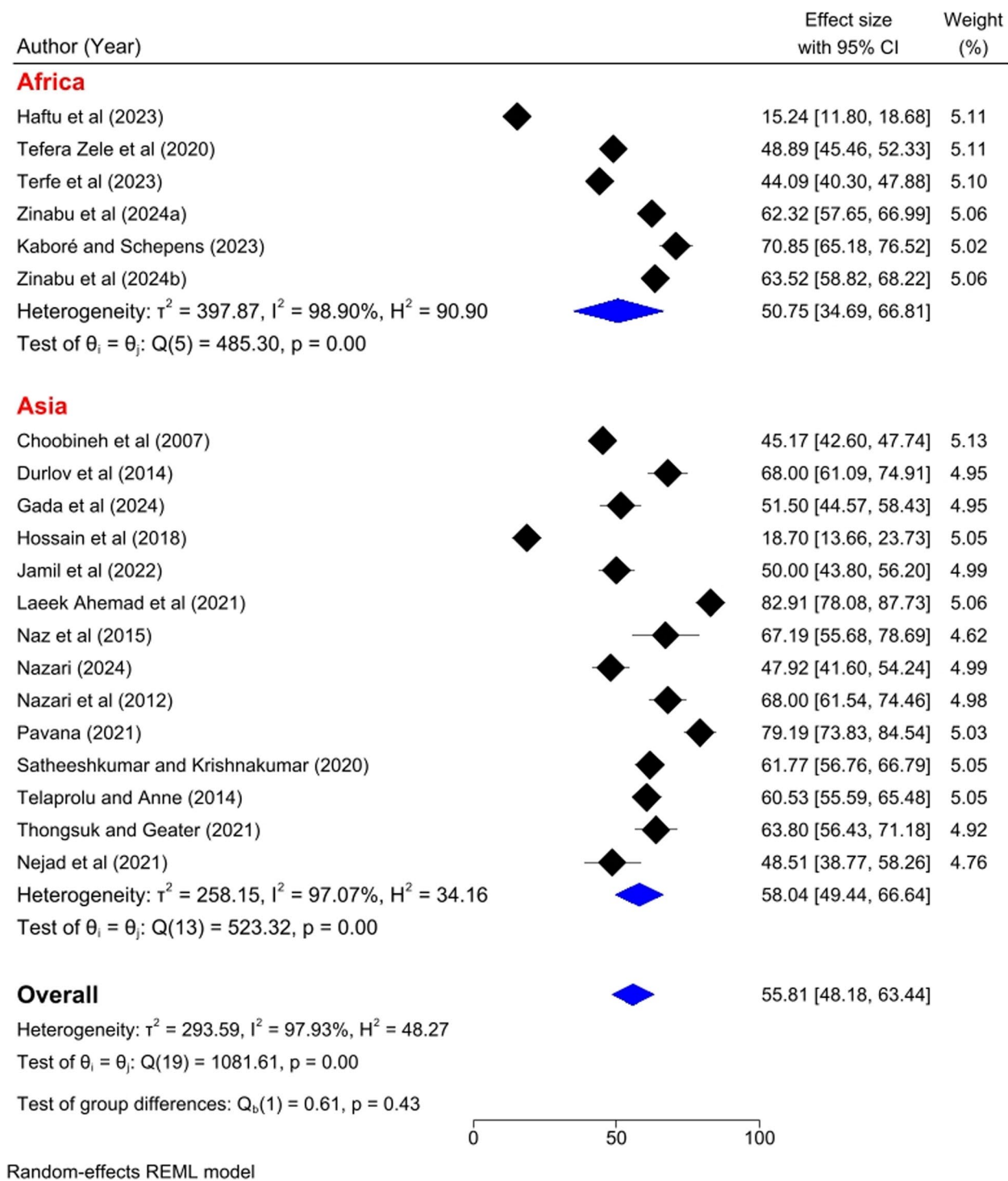


Fig. 5 Subgroup analysis by continent

and Jimma [78], respondents who performed their task in awkward postures/positions had a 3.59 and 4.09-fold higher possibility of developing WMSDs than those who did not, respectively.

The results of this meta-analysis revealed that the odds for the occurrence of LBP among weavers who had job

stress were 2.15 times higher as compared to those weavers who had no job stress. This result is congruent with the findings of the studies conducted in the Hunan province of China [80], and Switzerland [81]. The possible justification might be evidenced by the fact that work-stress-induced psychological burdens become heavier,

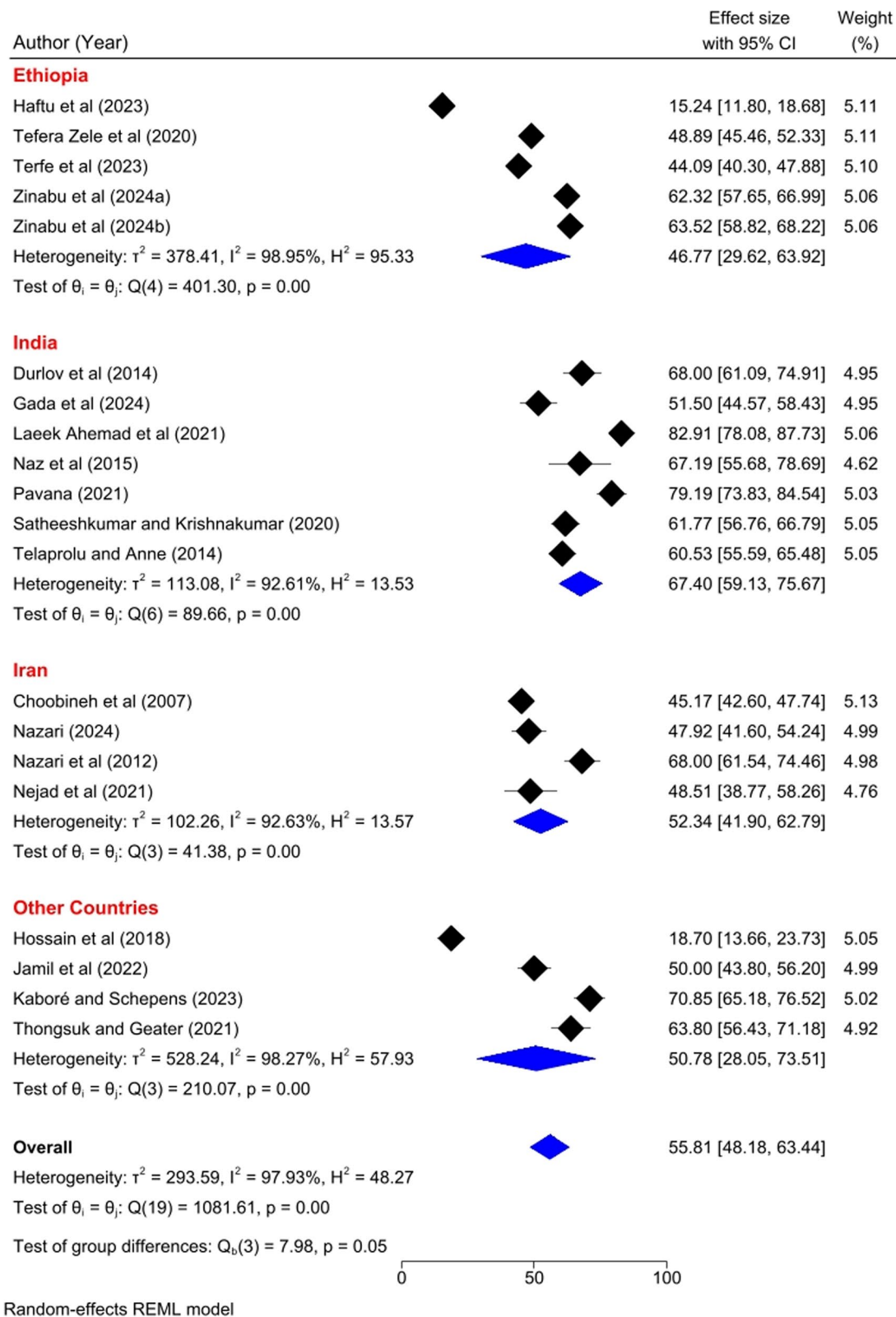


Fig. 6 Subgroup analysis by country category

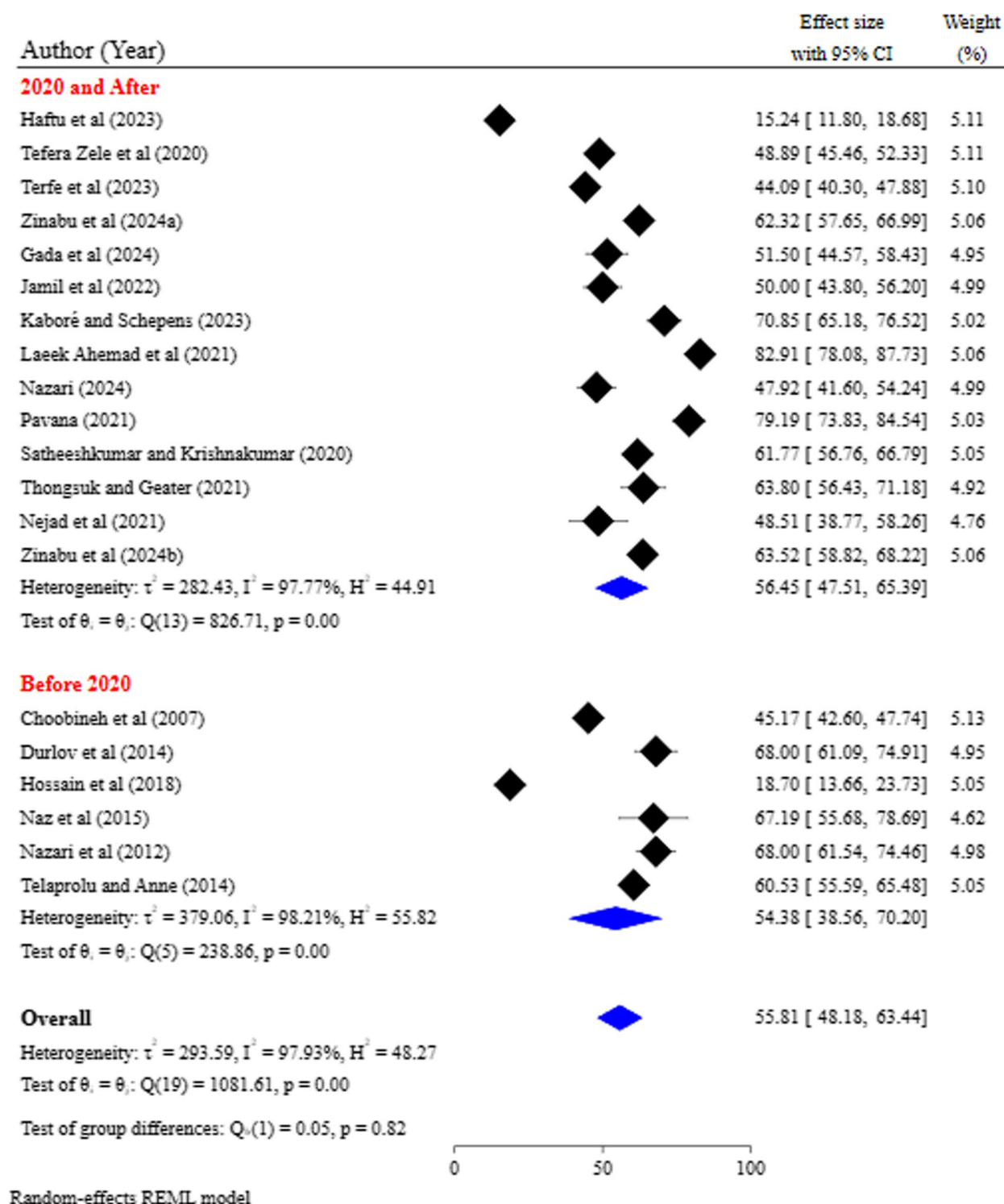


Fig. 7 Subgroup analysis by publication year category

which in turn intensifies muscle strain [82, 83]. Moreover, evidence showed that less satisfied workers with their job experienced more stress [84] and this in turn contributed to the high risk of developing WMSD as evidenced by the findings of the studies conducted in Nigeria [85]. Despite

this systematic review and meta-analysis following PRISMA protocols, drawing definitive conclusions about the determinants remains constrained by the cross-sectional design of the analyzed studies. Additionally, the study was restricted to manuscripts published in English,

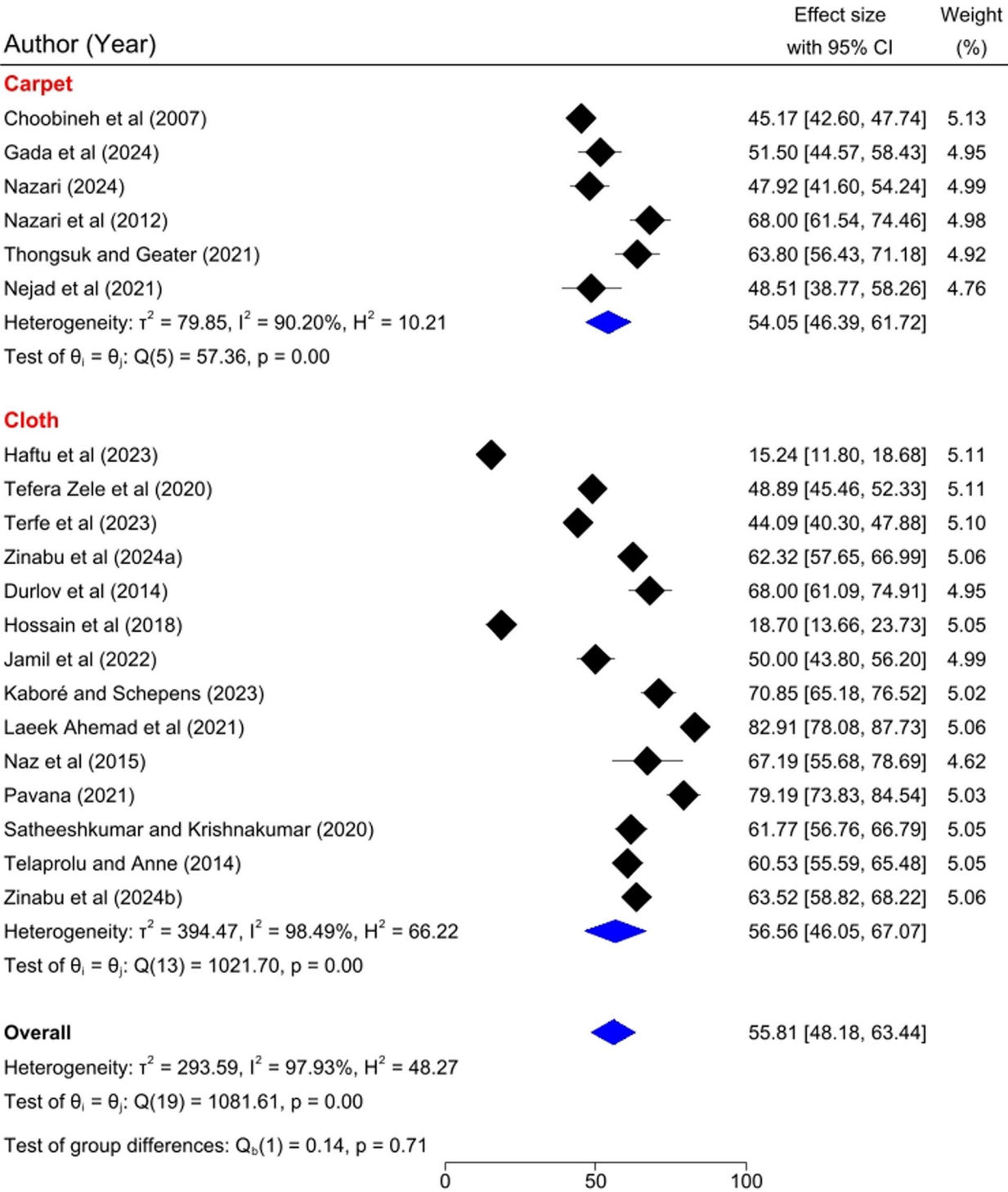


Fig. 8 Subgroup analysis by weaving type

potentially introducing reporting bias. Moreover, data from all LMICs were unavailable, thereby limiting the overall representativeness of the study. A substantial degree of variability among studies was observed, which may have impacted the applicability of our findings.

Although we conducted subgroup and meta-regression analyses that examine factors such as continent, country category, publication year category, and sample size category, weaving type, the method of data collection, none effectively accounted for the variability, indicating that

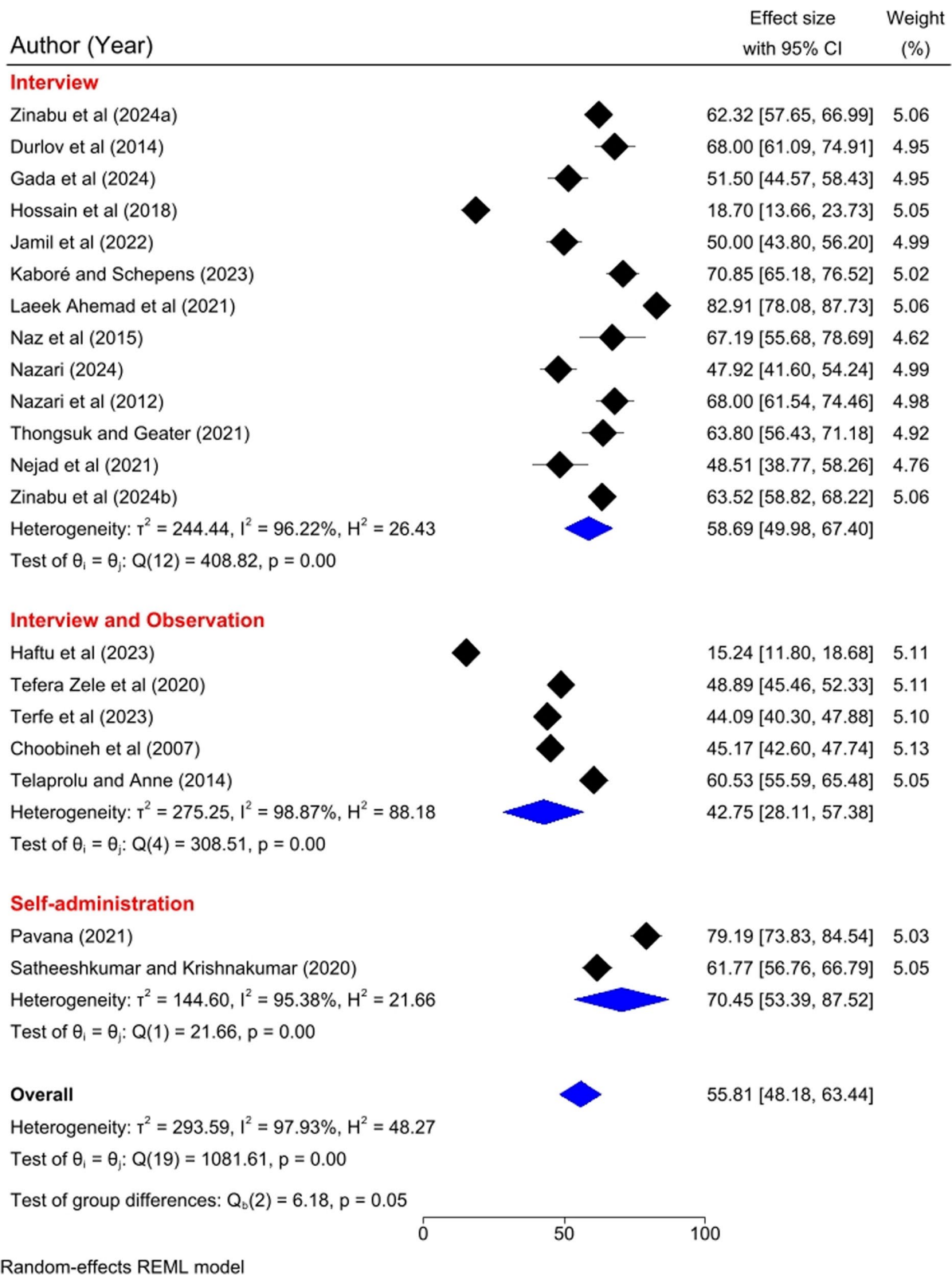


Fig. 9 Subgroup analysis by the method of data collection

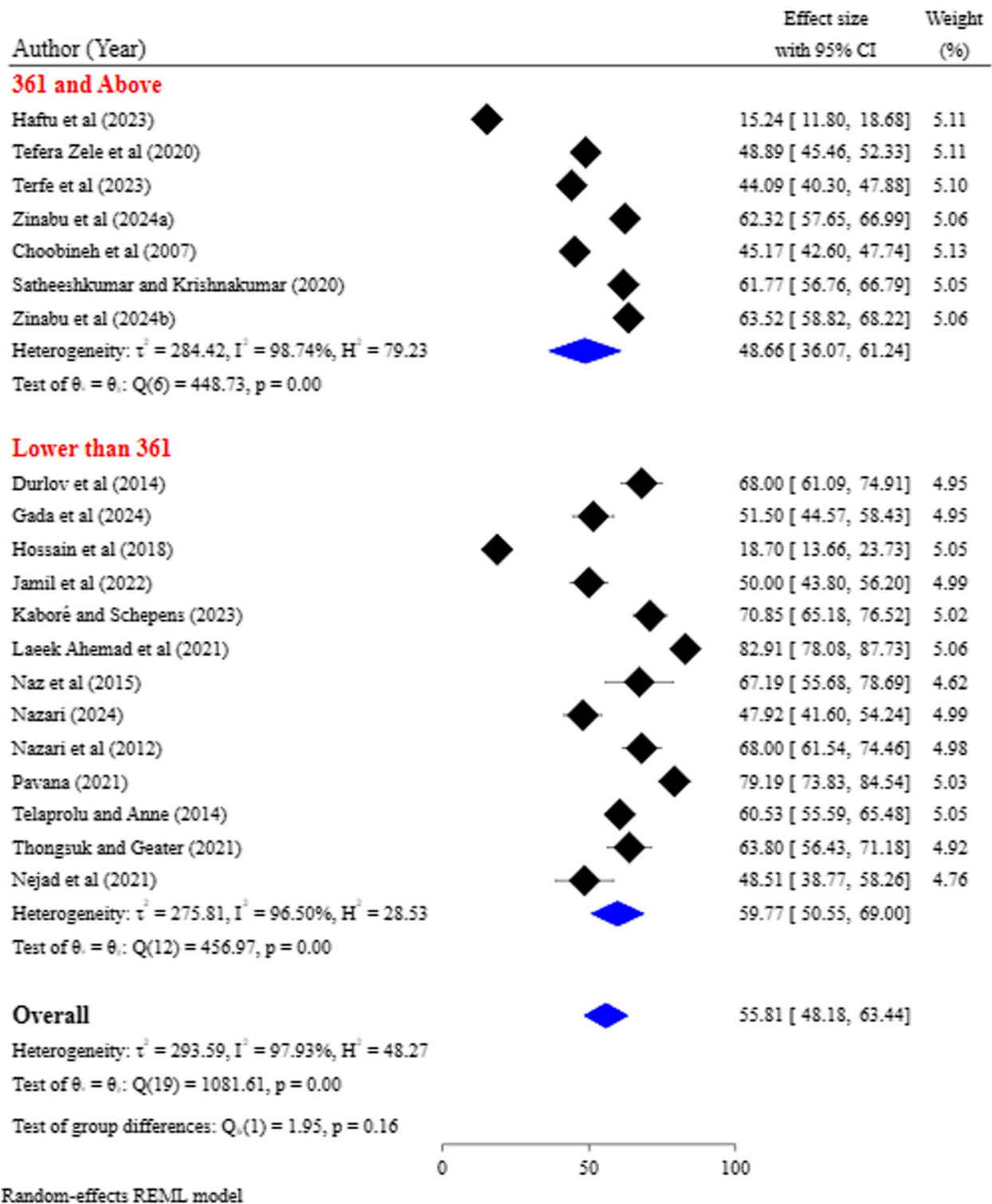


Fig. 10 Subgroup analysis by sample size category

Table 2 A univariate meta-regression analysis to pinpoint the factors associated with the heterogeneity of low back pain prevalence among weavers, 2024

Variables	Coefficient	p-value
Continent	−0.895285	0.956
Country category	6.638756	0.103
Publication year category	6.486906	0.515
Weaving type	0.6206141	0.956
Method of data collection	1.332248	0.868
Sample size category	15.11925	0.281

unidentified or unreported factors played a more substantial role.

Conclusion

In this study, the recorded pooled prevalence of LBP among weavers in low-and middle-income countries was higher. Age, working in a chair with no back support, working in an uncomfortable posture, work experience, and job stress were the factors significantly associated with LBP. This finding highlights significant health consequences, such as persistent discomfort, physical impairment, and a diminished standard of living. The repetitive movements and prolonged static postures involved in weaving contribute to widespread strain at the lower back region of the body, increasing the risk of long-term conditions such as arthritis and spine-related leg pain. Additionally, this study highlights the pressing necessity for policy actions and workplace improvements to safeguard weavers from enduring harm caused by LBP.

Moreover, the findings of this study provide critical insights for policy and clinical application, emphasizing the need for targeted interventions. Translating this evidence into practice involves integrating ergonomic guidelines such as introducing training sessions on proper posture, furnishing supportive seating, and improving workstation designs, and promoting accessible preventive healthcare measures, such as routine screenings and physical therapy. Clinicians and policymakers should collaborate to implement workplace adaptations that alleviate repetitive strain and prolonged static postures, thereby reducing the prevalence and severity of LBP. Additionally, public health strategies should prioritize education on proper posture and musculoskeletal health while advocating for mechanization to alleviate physical burden.

Performing physical exercise, implementing ergonomic design, provision of psychosocial training, reduction of job-related stresses, and improving job satisfaction and reduction of workload by the International Labor Organization (ILO), the World Health Organization (WHO), governments and health ministries of LMICs, weaving institution employers, and other partners and stakeholders are recommended in preventing and reducing

Table 3 The pooled odds ratio for factors associated with LBP among weavers in low-and middle-income countries, 2024

Listed variables	Number of study participants	Number of studies included	Pooled Odds Ratio (95% CI)	Heterogeneity I ²	p-value
Gender (female)	1799	4	1.04(0.70–1.37)	52.1%	0.100
Age > 40 Years	497	2	1.81(1.06–2.56)*	45.3%	0.176
Work experience > 10 Years	934	2	1.54(1.17–1.91)*	73.5%	0.052
Working hours per day (> 8 h)	650	2	1.48(0.83–2.14)	65.1%	0.091
Working in a chair with no back support (yes)	1217	2	2.20(1.54–2.87)*	0.0%	0.530
Working in an uncomfortable posture (yes)	778	2	2.57(1.25–3.90)*	0.0%	0.524
Job stress (yes)	1217	2	2.15(1.33–2.97)*	40.1%	0.196

*Shows significant association of variables with the pooled prevalence of LBP

work-related LBP in occupational area workers such as weavers.

Abbreviations

CI	Confidence Interval
DALYs	Disability-Adjusted Life Years
LBP	Low Back Pain
LMICs	Low-and Middle-Income Countries
MSDs	Musculoskeletal Disorders
POR	Pooled Odds Ratio
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analysis
WMSDs	Work-related Musculoskeletal Disorders
YLD	Years Lived with Disability

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12891-025-08967-4>.

Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Authors' contributions

AKG conceptualize and designed the study. AKG, CD and GB involved in the quality assessment of the studies. AKG, BD, GB, and LB, participated in the collection, extraction, analysis, and interpretation of the data, drafted, and prepared the manuscript. CD and BD assisted in the design and approved the manuscript with revisions. LB assisted in the design, approval of the manuscript with revisions, participated in data analysis, and revised the subsequent write-up of the paper. AKG also participated in the analysis and revision and writeup of the paper. All authors critically reviewed the draft manuscript and approved the final manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Competing interests

The authors declare no competing interests.

Conflict of interest

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